

Noetica Master Blueprint

SUMMARY

Scientific Significance Framework

PURPOSE

The purpose of the Scientific Significance Framework is to determine: - Which discoveries matter. - Why they matter. - How much they matter. - How confidently they can be evaluated.

Noetica does not rank papers. Noetica ranks discoveries. The Scientific Significance Framework is therefore the primary decision-making engine of the platform.

CORE PRINCIPLE

Traditional systems rank: - Citation count - Journal prestige - Popularity - Media attention

Noetica ranks: - Evidence - Significance - Influence - Connectivity - Future potential

PRIMARY QUESTION

How likely is this discovery to meaningfully advance human knowledge?

DISCOVERY-CENTRIC EVALUATION

Primary Entity: Discovery Not: Paper, Author, Journal, Organization Example: CRISPR is the discovery. Associated evidence: - papers - patents - grants - clinical trials - products - startups

The framework evaluates the discovery cluster as a whole.

RANKING LAYERS

Layer 1: Scientific Significance Layer 2: Technological Significance Layer 3: Civilizational Significance Layer 4: Future Potential

LAYER 1: SCIENTIFIC SIGNIFICANCE SCORE

Measures contribution to human knowledge. Components: - Evidence Strength (25%) - Novelty (15%) - Replication Confidence (15%) - Cross-Disciplinary Influence (10%) - Knowledge Connectivity (10%) - Citation Growth (10%) - Funding Momentum (10%) - Practical Impact (5%)

LAYER 2: TECHNOLOGICAL SIGNIFICANCE SCORE

Purpose: Measure ability to transform human capabilities. Components: - Patent Activity - Industry Adoption - Open Source Adoption - Infrastructure Impact - Commercialization Potential - Technology Diffusion Examples: Transistor, Internet, GPS, CRISPR, AlphaFold, Transformers

LAYER 3: CIVILIZATIONAL SIGNIFICANCE SCORE

Purpose: Measure impact on civilization. Components: - Health - Education - Economy - Science - Technology - Energy - Governance - Human Capability Examples: Calculus, Evolution, Relativity, DNA, Internet, Human Genome

LAYER 4: FUTURE POTENTIAL SCORE

Purpose: Estimate future influence. Questions: - Could this become foundational? - Could this create a new field? - Could this transform existing fields? - Could this enable future discoveries? Forecast Outputs: - Probability - Confidence Interval - Risk Assessment - Alternative Scenarios No certainty claims are allowed.

SOCIAL MEDIA POLICY

Social Media Weight: 0% Scientific importance and public attention are different concepts. Allowed: - Trend Detection - Weak Signal Detection - Emerging Community Observation Not Allowed For: - Ranking - Significance Scoring - Evidence Assessment - Civilizational Impact - Scientific Credibility

SOURCE RELIABILITY ENGINE

Every source receives a confidence score. Tier 1 (Highest Reliability): Nature, Science, Cell, NEJM, Major Scientific Organizations Tier 2 (Strong Reliability): Peer-reviewed journals Tier 3 (Moderate Reliability): Preprints (arXiv, bioRxiv, medRxiv) Tier 4 (Weak Reliability): Blogs, Media Articles, Opinion Pieces Tier 5 (Minimal Reliability): Social Media, Unverified Claims, Rumors

DISCOVERY LIFECYCLE SCORING

Speculative → Emerging → Growing → Breakthrough → Established → Foundational → Civilizational → Historical

DISCOVERY MOMENTUM ENGINE

Measures: - Research Growth - Funding Growth - Patent Growth - Dataset Growth - Open Source Growth - Research Community Growth

DISCOVERY CONFIDENCE SCORE

Measures: - Evidence Quality - Replication Strength - Source Reliability - Consensus Level

SCIENTIFIC CONSENSUS SCORE

Measures agreement among experts.

SCIENTIFIC DEBATE SCORE

Measures degree of active disagreement.

WEAK SIGNAL DETECTION

Purpose: Identify important discoveries before widespread recognition. Examples: - Transformers (2017) - CRISPR (early years) - AlphaFold (before adoption) - Protein Language Models

RANKING TRANSPARENCY REQUIREMENT

Every ranking must explain itself. Example: Discovery: Protein Language Models Scientific Significance: 91.4 Evidence: 9.2 Novelty: 8.9 Connectivity: 9.5 Funding: 8.4 Cross-Field Influence: 9.7 Users must always understand why a discovery received a score.

FINAL PRINCIPLE

The objective of the Scientific Significance Framework is not to identify what is popular. The objective is to identify what matters. Noetica therefore treats scientific intelligence as a problem of evidence, influence, connectivity, and long-term significance rather than attention, engagement, or visibility.